Tool	DevOps phase	Tool type	Configuration format
GitHub	Continuous development	Source code management	UI
GitLab	Continuous development	Source code management	UI
Gradle	Continuous development	Build	Groovy
Maven	Continuous development	Build	XML
Jira	Continuous development	Project management	UI
Kanban	Continuous development	Project management	UI
Jenkins	Continuous integration	Build, testing, delivery	UI
Bamboo	Continuous integration	Build	UI
JUnit	Continuous testing	Unit testing	Java
Selenium	Continuous testing	Web testing	Python, Java
Puppet	Continuous deployment	Configuration management	Domain specific language
Ansible	Continuous deployment	Configuration management	YAML
Docker	Continuous deployment	Containerization	YAML
Chef	Continuous deployment	Configuration management	Ruby-domain specific language
ELK Stack	Continuous monitoring	Monitoring	Domain specific language
Nagios	Continuous monitoring	Network monitoring	-
Slack	Continuous feedback	Collaborative software	UI
Kubernetes	Continuous operations	Orchestration	YAML
OpenStack	Continuous operations	IaaS	UI

Table 1: Automation tools of DevOps

tools commonly used to achieve automation that specifically put emphasis on continuous integration, continuous delivery, or continuous deployment. Incident monitoring can be done in real-time enabling quick rollback methods. Reliability and traffic handling can be achieved through the implementation of cloud computing and microservices. All these establish the fact that DevOps can deliver what it promises.

Phases and toolchains

Quality deliveries with short cycle times require a high degree of automation. Automation is necessary to achieve DevOps as it leads to higher deliverables and optimises the deployed application services. The DevOps life cycle is achieved through the continuous completion of various phases:

- Continuous development
- Continuous integration
- Continuous testing
- Continuous deployment
- Continuous feedback
- Continuous monitoring
- Continuous operations

Everything is continuous in DevOps, and many of these phases interact with one another and work in a pipeline to assemble an application to achieve the organisation's goals. These phases are fulfilled with various toolchains of selection; some of the common tools are mentioned in Table 1.

Continuous development

The first step in the DevOps pipeline is to store the development code on a version control system (VCS) where the development team can collaborate on the code and manage different versions of the development. Team members may have their own set of source codes in a project depending on the module they are working on. To handle these versions and integrate them into one project, a source code manager is required.

Git, an open source version control software, efficiently manages various versions of the source code and coordinates changes in the files made by collaborating developers. It speeds up data integrity and code collaboration within the team. The workflow of GitHub is showcased in Figure 1.

There are several source code holding facilities that utilise the Git